MC210401814: IQRA ZAMEER

Time Left 86 sec(s)

MTH603 - Numerical Analysis (Quiz No.3)

Quiz Start Time: 06:29 PM, 13 August 2022

Question # 1 of 10 (Start time: 06:29:45 PM, 13 August 2022)

Total Marks: 1

For the given data points (x_0, y_0), $(x_1 y_1)$, $(x_2 y_2)$, and (x_3, y_3) the zero − order divide difference will be given as

Select the correct option

Reload Math Equations

○	$y[y_0]$
○	$y[x_0]$
○	$y[y_0 . y_1]$
○	$y[x_0 . x_1]$

Click to Save Answer & Move to Next Question

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Question # 2 of 10 (Start time: 06:10:51 PM, 13 August 2022) **Total Marks: 1**

there are (n+2) values of y corresponding to (n+2) values of x, then we can represent the function f(x) by a polynomial of degree

Select the correct option

Reload Math Equations

○	n
○	n+1
○	n+2
○	n-1

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Time Left 88

MTH603 - Numerical Analysis (Quiz No.3)

Quiz Start Time: 05:31 PM, 13 August 2022

Question # 7 of 10 (Start time: 05:38:32 PM, 13 August 2022)

Total Marks: 1

Let

$$f(x,y) = 2x^3 + 6y^3 + 9xy$$

For x=0,1,2,3,4, and y=0, 1,2,3,4 Then computing the value of f(2.5,3.5) is an example of ....

Select the correct option

Reload Math Equations

- Interpolation in one dimension
- Interpolation in two dimensions
- Interpolation in four dimensions
- Interpolation in three dimensions

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For the given data points $(4, 1.3)$, $(8, 1.5)$, *and* $(12, 1.9)$ *the divide difference table will be given as*

the correct option

Reload Math Equ

x	y	$1stD.D$	$2ndD.D$
4	1.3	0.1	
8	1.5	0.35	0.0062
12	1.9		

x	y	$1stD.D$	$2ndD.D$
4	1.3	0.05	
8	1.5	0.1	0.0062
12	1.9		

x	y	$1stD.D$	$2ndD.D$
4	1.3	0.0062	
8	1.5	0.1	0.05
12	1.9		

x	y	$1stD.D$	$2ndD.D$
4	1.3	0.1	
8	1.5	0.0062	0.05
12	1.9		

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Question # 4 of 10 (Start time: 06:12:00 PM, 13 August 2022)

Differential operator in terms of forward difference operator is given by

Select the correct option

- $D = \frac{1}{h}(\Delta - \frac{\Delta^2}{2!} + \frac{\Delta^3}{3!} - \frac{\Delta^4}{4!} + \frac{\Delta^5}{5!} - \dots)$
- $D = \frac{1}{h}(\Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - \frac{\Delta^4}{4} + \frac{\Delta^5}{5} - \dots)$
- $D = \frac{1}{h}(\Delta + \frac{\Delta^2}{2} + \frac{\Delta^3}{3} + \frac{\Delta^4}{4} + \frac{\Delta^5}{5} + \dots)$
- $D = \frac{1}{h}(\Delta + \frac{\Delta^2}{2!} + \frac{\Delta^3}{3!} + \frac{\Delta^4}{4!} + \frac{\Delta^5}{5!} + \dots)$

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Question # 10 of 10 (Start time: 05:46:50 PM, 13 August 2022)

What will be the value of 'a' in the given divide difference table?

x	y	$1st D.D$	$2nd D.D$
3	0.4		
6	0.9	0.1667	
9	1.7	0.2667	a

Select the correct option

- 0.0271
- 0.0967
- 0.0349
- 0.0254

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Question # 4 of 10 (Start time: 12:04:31 PM, 11 August 2022)

If any or all of the artificial variables do not leave the basis in the final solution, then this indicates that the problem _____

Select the correct option

- must have a solution
- does not have a solution
- may have a solution
- non of these

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Question # 1 of 10 (Start time: 06:10:03 PM, 13 August 2022)

What will be the value of 'a' in the given divide difference table?

x	y	$1stD.D$	$2ndD.D$	$3rdD.D$
2	0.5	0.3	a	
4	1.1	0.3	-0.0125	-0.0021
6	1.7	0.25		
8	2..2			

Select the correct option

- 0.0893
- 0.0612
- 0.0115
- 0

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MTH603 - Numerical Analysis (Quiz No.3)

Question # 10 of 10 (Start time: 06:28:34 PM, 13 August 2022)

If $f(x) = 2x^3 - 5x^2 + 9x - 6$, then its------derivative is zero for all x.

Select the correct option

- 3rd
- 5th
- 2nd
- 4th

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MTH603 - Numerical Analysis (Quiz No.3)

Question # 8 of 10 (Start time: 06:36:10 PM, 13 August 2022)

Zero-th order divided difference is defined as

Select the correct option

- $y[x_0]=y_0$
- None of the given choices
- $y[x_0]=x_0$
- $y[x_0]=y_1$

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Question # 9 of 10 (Start time: 05:38:58 PM, 13 August 2022)

: What will be the value of first order divided difference f[1,5]for the following data x:0 1 5 y:2 1 5

Select the correct option

- 0
- 3
- 2
- 1

Question # 9 of 10 (Start time: 06:28:01 PM, 13 August 2022)

$\delta = - - --$

Select the correct option

- $E^{\frac{1}{2}} + E^{-\frac{1}{2}}$
- $E^{\frac{1}{2}} - E^{-\frac{1}{2}}$
- $\frac{E^{\frac{1}{2}} + E^{-\frac{1}{2}}}{2}$
- None

Question # 3 of 10 (Start time: 06:11:28 PM, 13 August 2022)

Zero-th order divided difference is defined as

Select the correct option

- ○ y[x0]=y0
- ○ None of the given choices
- ○ y[x0]=x0
- ○ y[x0]=y1

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Question # 6 of 10 (Start time: 06:25:06 PM, 13 August 2022)

Let

$$f(x,y) = 2x^3 + 6y^3 + 9xy$$

For x=0, 1,2,3,4, and y=0, 1,2,3,4 Then computing the value of f(2.5,3.5) is an example of .....

Select the correct option

○	Interpolation in four dimensions
○	Interpolation in three dimensions
○	Interpolation in one dimension
○	Interpolation in two dimensions

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al Analysis (Quiz No.3)

Quiz Start Time: 06:21 PM, 13 Au

Start time: 06:27:06 PM, 13 August 2022)

Tot

If any three data points *are given, the formula for Lagrange's* interpolation *polynomial will be*

ption

Reload Math Ec

$$y = f(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} y_0 + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} y_1 + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} y_2$$

$$y = f(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} y_2 + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} y_1 + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} y_0$$

$$y = f(x) = \frac{(x - x_1)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} y_0 + \frac{(x - x_0)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} y_1 + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} y_2$$

$$y = f(x) = \frac{(x_0 - x_1)(x_0 - x_2)}{(x - x_1)(x - x_2)} y_0 + \frac{(x_1 - x_0)(x_1 - x_2)}{(x - x_0)(x - x_2)} y_1 + \frac{(x_2 - x_0)(x_2 - x_1)}{(x - x_0)(x - x_1)} y_2$$

Click to Save Answer & Move to Next Qu

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For the given data points (x_0, y_0), $(x_1 y_1)$, $(x_2 y_2)$, and (x_3, y_3) the first − order divide difference will be given as

e correct option

Reload Math Equati

$y[x_0]$

$y[y_0, y_1, y_2]$

$y[x_0, x_1]$

$y[x_0, x_1, x_2]$

Click to Save Answer & Move to Next Questi

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Question # 9 of 10 (Start time: 06:17:30 PM, 13 August 2022) **Total Marks:** 1

If the given tabular function f(x) is approximated by the polynomial 'P1(x) = x+1' then which of the following polynomial will approximate the derivative of f(x) ?

Select the correct option

- 1
- 2x
- x-1
- x

Click to Save Answer & Move to Next Question

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Question # 1 of 10 (Start time: 06:21:30 PM, 13 August 2022)

Lagrange's interpolation formula is used when the values of the independent variable are

Select the correct option

- ○ Not equally spaced
- ○ None
- ○ Constant
- ○ Equally spaced

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Question # 4 of 10 (Start time: 06:23:40 PM, 13 August 2022)

If $f(x) = 2x$, then which of the following is will be derivative of $f(x)$ at $x = 0.2$?

Select the correct option

- 0.2
- 0.4
- -2
- 2

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Question # 5 of 10 (start time: 06:24:22 PM, 13 August 2022)

For the given data points $(4, 45)$, $(5, 104)$, *and* $(6, 190)$, *the* zero $-$ *order divide difference will be*

Select the correct option

Reload

○	42
○	none
○	35
○	46

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Question # 10 of 10 (Start time: 06:18:23 PM, 13 August 2022) **Total Marks:** 1

For the giev three data points, *the* degree *of Lagrange's* interpolation *polynomial could be*

x	0.3	0.7	0.9
y	0.067	0.248	0.518

Select the correct option

Reload Math Equations

- $y = f(x) = \frac{(x-0.7)(x-0.9)}{(0.3-0.7)(0.3-0.9)}(0.518) + \frac{(x-0.3)(x-0.9)}{(0.7-0.3)(0.7-0.9)}(0.248) + \frac{(x-0.3)(x-0.7)}{(0.9-0.3)(0.9-0.7)}(0.067)$
- $y = f(x) = \frac{(x-0.7)(x-0.9)}{(0.3-0.7)(0.3-0.9)}(0.067) + \frac{(x-0.3)(x-0.9)}{(0.7-0.3)(0.7-0.9)}(0.248) + \frac{(x-0.3)(x-0.7)}{(0.9-0.3)(0.9-0.7)}(0.518)$
- $y = f(x) = \frac{(x-0.7)(x-0.9)}{(0.3-0.7)(0.3-0.9)}(0.248) + \frac{(x-0.3)(x-0.9)}{(0.7-0.3)(0.7-0.9)}(0.067) + \frac{(x-0.3)(x-0.7)}{(0.9-0.3)(0.9-0.7)}(0.518)$
- $y = f(x) = \frac{(x-0.7)(x-0.9)}{(0.3-0.7)(0.3-0.9)}(0.067) + \frac{(x-0.3)(x-0.9)}{(0.7-0.3)(0.7-0.9)}(0.518) + \frac{(x-0.3)(x-0.7)}{(0.9-0.3)(0.9-0.7)}(0.248)$

CamScanner

Which of the following is the Richardson's Extrapolation limit: F1(h/2) provided that F(h/2) = F(h) = 1 ?

Select the correct option

- 1
- 0
- 3
- 4

For the given data points $(4, 2.2)$, $(8, 3.5)$, *and* $(12, 4.1)$ *the divide difference table will be given as*

e correct option

Reload Math

x	y	$1st D.D$	$2nd D.D$
4	2.2	0.325	
8	3.5	0.15	-0.065
12	4.1		

x	y	$1st D.D$	$2nd D.D$
4	2.2	0.325	
8	3.5	0.15	-0.098
12	4.1		

x	y	$1st D.D$	$2nd D.D$
4	2.2	0.325	
8	3.5	0.15	-0.0108
12	4.1		

x	y	$1st D.D$	$2nd D.D$
4	2.2	0.325	
8	3.5	0.15	-0.0219
12	4.1		

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Question # 7 of 10 (Start time: 06:26:21 PM, 13 August 2022)

1st ordered divided difference formula is defined as

Select the correct option

- $y[x0,x1]=(y1+y0)/(x1-x0)$
- $y[x0,x1]=(y1-y0)/(x1+x0)$
- None of the given choices
- $y[x0,x1]=(y1-y0)/(x1-x0)$

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Question # 7 of 10 (Start time: 06:15:12 PM, 13 August 2022)

Given the following data x: 1 3 -7 f(x): 3 -6 -2 Which formula is useful in finding the interpolating polynomial?

Select the correct option

○	Newton's backward difference formula
○	Newton's forward difference formula
○	Lagrange's interpolation formula
○	None of the given choices

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Question # 2 of 10 (start time: 05:10:37 PM, 13 August 2022) **Total Marks:** 1

If only two data points *are given, the formula for Lagrange's* interpolation *polynomial will be*

Select the correct option

Reload Math Equations

- $y = f(x) = \frac{(x - x_0)}{(x_0 - x_1)} y_0 + \frac{(x - x_1)}{(x_1 - x_0)} y_1$
- $y = f(x) = \frac{(x_1 - x_0)}{(x - x_0)} y_0 + \frac{(x_0 - x_1)}{(x - x_1)} y_1$
- $y = f(x) = \frac{(x - x_1)}{(x_0 - x_1)} y_0 + \frac{(x - x_0)}{(x_1 - x_0)} y_1$
- $y = f(x) = \frac{(x - x_0)}{(x_1 - x_0)} y_0 + \frac{(x - x_1)}{(x_0 - x_1)} y_1$

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Question # 3 of 10 (Start time: 06:23:02 PM, 13 August 2022) **Total Marks:** 1

For the giev three data points, *the* degree *of Lagrange's* interpolation *polynomial could be*

x	0.3	0.7	0.9
y	0.067	0.248	0.518

Select the correct option

Reload Math Equations

- $y = f(x) = \frac{(x-0.7)(x-0.9)}{(0.3-0.7)(0.3-0.9)}(0.067) + \frac{(x-0.3)(x-0.9)}{(0.7-0.3)(0.7-0.9)}(0.518) + \frac{(x-0.3)(x-0.7)}{(0.9-0.3)(0.9-0.7)}(0.248)$
- $y = f(x) = \frac{(x-0.7)(x-0.9)}{(0.3-0.7)(0.3-0.9)}(0.518) + \frac{(x-0.3)(x-0.9)}{(0.7-0.3)(0.7-0.9)}(0.248) + \frac{(x-0.3)(x-0.7)}{(0.9-0.3)(0.9-0.7)}(0.067)$
- $y = f(x) = \frac{(x-0.7)(x-0.9)}{(0.3-0.7)(0.3-0.9)}(0.248) + \frac{(x-0.3)(x-0.9)}{(0.7-0.3)(0.7-0.9)}(0.067) + \frac{(x-0.3)(x-0.7)}{(0.9-0.3)(0.9-0.7)}(0.518)$
- $y = f(x) = \frac{(x-0.7)(x-0.9)}{(0.3-0.7)(0.3-0.9)}(0.067) + \frac{(x-0.3)(x-0.9)}{(0.7-0.3)(0.7-0.9)}(0.248) + \frac{(x-0.3)(x-0.7)}{(0.9-0.3)(0.9-0.7)}(0.518)$

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Question # 10 of 10 (start time: 05:18:46 PM, 13 August 2022)

x: 1 3 7 10 f(x): −3 10 3 13 Which formula is useful in finding the interpolating polynomial?

Select the correct option

- ○ Lagrange's interpolation formula
- ○ Newton's backward difference formula
- ○ Newton's forward difference formula
- ○ (viii) None of the given choices

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Question # 6 of 10 (Start time: 04:51:51 PM, 13 August 2022)

Given the following data x1 2 7 11 y5 10 13 37 Which formula is useful in finding the interpolating polynomial?

Select the correct option

- Newton's forward difference interpolation formula
- None
- Newton's backward difference interpolation formula
- Lagrange's interpolation formula

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Question # 8 of 10 (Start time: 04:53:08 PM, 13 August 2022)

In Lagrange's interpolation, for the given five points we can represent the function f (x) by a polynomial of degree

Select the correct option

- 5
- 4
- 3
- 6

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Question # 5 of 10 (start time: 05:13:58 PM, 13 August 2022)

If $f(x) = 2x^3 - 5x^2 + 9x - 6$, then its------derivative is zero for all x.

Select the correct option

○	3rd
○	5th
○	4th
○	2nd

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Question # 5 of 10 (Start time: 04:53:48 PM, 13 August 2022)

x: 1 3 7 f(x): 1 4 9 f(3) Can be found using

Select the correct option

- None of the given choices
- Newton's backward difference formula
- Newton's forward difference formula
- Lagrange's interpolation formula

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Question # 6 of 10 (Start time: 05:14:32 PM, 13 August 2022) **Total Marks:** 1

For a function 'f(x) = x', with a step size of 'h=0.01', which of the following gives the 1st derivative at x =1 by using two point formula?

Select the correct option

- y'(x) = 0.1 + Some Truncation Error
- y'(x) = -0.1 + Some Truncation Error
- y'(x) = 1+Some Truncation Error
- y'(x) = 1.01 + Some Truncation Error

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Question # 2 of 10 (Start time: 04:46:03 PM, 13 August 2022)

Which of the following is the Richardson's Extrapolation limit: F3(h/8) provided that F2(h/8) = F2(h/4) = 1 ?

Select the correct option

- 1
- -1
- 63
- 64

Question # 3 of 10 (Start time: 05:11:41 PM, 13 August 2022) **Total Marks:**

For the given divide difference table

x	y	$1stD.D$	$2ndD.D$
1	2.2	0.4333	
4	3.5	0.2	−0.0389
7	4.1		

the Newton's divide difference interpolation *formula will be*

Select the correct option

Reload Math Equations

- $y = f(x) = -0.0389 + (x - 1)(2.2) + (x - 1)((x - 4)(0.4333)$
- $y = f(x) = -0.0389 + (x - 1)(0.4333) + (x - 1)((x - 4)(2.2)$
- $y = f(x) = 2.2 + (x - 1)(-0.0389) + (x - 1)((x - 4)(0.4333)$
- $y = f(x) = 2.2 + (x - 1)(0.4333) + (x - 1)((x - 4)(-0.0389)$

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Question # 4 of 10 (Start time: 04:49:49 PM, 13 August 2022)

For the given data points $(1, -3)$, $(2, 0)$, *and* $(3, 15)$, *the zero − order divide difference will be*

Select the correct option

Reload Matl

- -3
- -2
- 0
- -1

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Question # 4 of 10 (Start time: 04:49:49 PM, 13 August 2022)

For the given data points (1, −3), (2, 0), and (3, 15), the zero − order divide difference will be

Select the correct option

Reload Matr

- -2
- -2
- 0
- -1

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Question # 7 of 10 (Start time: 04:52:35 PM, 13 August 2022)

Given the following data x: 1 3 -7 f(x): 3 -6 -2 Which formula is useful in finding the interpolating polynomial?

Select the correct option

- Newton's forward difference formula
- Newton's backward difference formula
- None of the given choices
- Lagrange's interpolation formula

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Question # 1 of 10 (start time: 05:09:29 PM, 13 August 2022) **Total Marks:** 1

For the given divide difference table

x	y	$1stD.D$	$2ndD.D$
2	2.2	0.325	
6	3.5	0.15	−0.0219
10	4.1		

the Newton's divide difference interpolation *formula will be*

Select the correct option

Reload Math Equations

○	$y = f(x) = 2.2 + (x-2)((x-6)(0.325) + (x-2)(-0.0219)$
○	$y = f(x) = 2.2 + (x-2)(0.325) + (x-2)((x-6)(-0.0219)$
○	$y = f(x) = 2.2 + (x-12)(0.325) + (x-6)((x-2)(-0.0219)$
○	$y = f(x) = 2.2 + (x-6)(0.325) + (x-12)((x-6)(-0.0219)$

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Question # 5 of 10 (Start time: 04:50:33 PM, 13 August 2022)

Total Marks:

For the given data points (1, 0.3), (3, 1), and (5, 1.2) the divide difference table will be given as

Select the correct option

Select MCQ's Operations

x	y	$1^{st}D.D$	$2^{nd}D.D$
2	0.3	0.35	
4	1	0.1	-0.225
6	1.2		

x	y	$1^{st}D.D$	$2^{nd}D.D$
2	0.3	0.35	
4	1	0.1	-0.325
6	1.2		

x	y	$1^{st}D.D$	$2^{nd}D.D$
2	0.3	0.35	
4	1	0.1	-0.125
6	1.2		

x	y	$1^{st}D.D$	$2^{nd}D.D$
2	0.3	0.35	
4	1	0.1	-0.0625
6	1.2		

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If any ten data points are given,the degree of Lagrange's interpolation polynomial could be

Select the correct option

- twelve
- eleven
- ten
- nine

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Question # 1 of 10 (Start time: 04:47:23 PM, 13 August 2022)

Lagrange's interpolation formula is used when the values of the independent variable are

Select the correct option

- None
- Equally spaced
- Not equally spaced
- Constant

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Question # 7 of 10 (Start time: 04:41:55 PM, 13 August 2022) **Total Marks: 1**

What will be the value of 'a' in the given divide difference table?

x	y	$1st D.D$	$2nd D.D$
3	0.4		
6	0.9	0.1667	
9	1.7	0.2667	a

Select the correct option

Reload Math Equations

- 0.0349
- 0.0254
- 0.029
- 0.0167

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MTH603 - Numerical Analysis (Quiz No.3)

Quiz Start Time: 04:02 PM, 13 August 202

Question # 6 of 10 (Start time: 04:07:27 PM, 13 August 2022)

Total Marks:

For the given data points $(4, 2.2)$, $(8, 3.5)$, *and* $(12, 4.1)$ *the divide difference table will be given as*

Select the correct option

x	y	$1st D.D$	$2nd D.D$
4	2.2	0.325	
8	3.5	0.15	-0.0108
12	4.1		

x	y	$1st D.D$	$2nd D.D$
4	2.2	0.325	
8	3.5	0.15	-0.0219
12	4.1		

x	y	$1st D.D$	$2nd D.D$
4	2.2	0.325	
8	3.5	0.15	-0.065
12	4.1		

x	y	$1st D.D$	$2nd D.D$
4	2.2	0.325	
8	3.5	0.15	-0.098
12	4.1		

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Question # 9 of 10 (Start time: 05:17:55 PM, 13 August 2022)

Differential operator in terms of forward difference operator is given by

Select the correct option

- ○ $D = \frac{1}{h}(\Delta - \frac{\Delta^2}{2!} + \frac{\Delta^3}{3!} - \frac{\Delta^4}{4!} + \frac{\Delta^5}{5!} - \dots)$
- ○ $D = \frac{1}{h}(\Delta + \frac{\Delta^2}{2!} + \frac{\Delta^3}{3!} + \frac{\Delta^4}{4!} + \frac{\Delta^5}{5!} + \dots)$
- ○ $D = \frac{1}{h}(\Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - \frac{\Delta^4}{4} + \frac{\Delta^5}{5} - \dots)$
- ○ $D = \frac{1}{h}(\Delta + \frac{\Delta^2}{2} + \frac{\Delta^3}{3} + \frac{\Delta^4}{4} + \frac{\Delta^5}{5} + \dots)$

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Question # 6 of 10 (start time: 04:40:50 PM, 13 August 2022) **Total Marks:** 1

If only two data points *are given, the formula for Lagrange's* interpolation *polynomial will be*

Select the correct option

Reload Math Equations

- $y = f(x) = \frac{(x - x_0)}{(x_1 - x_0)} y_0 + \frac{(x - x_1)}{(x_0 - x_1)} y_1$
- $y = f(x) = \frac{(x_1 - x_0)}{(x - x_0)} y_0 + \frac{(x_0 - x_1)}{(x - x_1)} y_1$
- $y = f(x) = \frac{(x - x_0)}{(x_0 - x_1)} y_0 + \frac{(x - x_1)}{(x_1 - x_0)} y_1$
- $y = f(x) = \frac{(x - x_1)}{(x_0 - x_1)} y_0 + \frac{(x - x_0)}{(x_1 - x_0)} y_1$

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MTH603 - Numerical Analysis (Quiz No.3)

Quiz Start Time: 05:09 PM, 13 August 2022

Question # 8 of 10 (Start time: 05:17:18 PM, 13 August 2022)

Total Marks: 1

For the given data points (x_0, y_0), $(x_1 y_1)$, $(x_2 y_2)$, and (x_3, y_3) the first − order divide difference will be given as

Select the correct option

Reload Math Equations

- $y[x_0, x_1]$
- $y[x_0]$
- $y[y_0, y_1, y_2]$
- $y[x_0, x_1, x_2]$

Click to Save Answer & Move to Next Question

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Question # 8 of 10 (Start time: 04:43:01 PM, 13 August 2022)

Given the following data x:4 5 7 10 y:46 102 294 346 Value of 1st order divided difference f[5 , 7] is

Select the correct option

○	91
○	92
○	96
○	94

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Question # 1 of 10 (Start time: 04:36:37 PM, 13 August 2022)

Given the following data x:1 3 8 y:2 4 9 f(3) can be found by using

Select the correct option

- Newton's backward difference interpolation formula
- Lagrange's interpolation formula
- Newton's forward difference interpolation formula
- None

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Question # 9 of 10 (start time: 04:44:50 PM, 13 August 2022) **Total Marks:** 1

For the given data points $(2,5)$, $(4,7)$, *and* $(6,9)$, *the zero − order divide difference will be*

Select the correct option

Reload Math Equations

- 2
- 1
- 5
- 0

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Question # 5 of 10 (Start time: 04:39:52 PM, 13 August 2022)

Which of the following method can be used for interpolation *for the given values of* x *and* y*?*

x	0.3	0.7	0.9
y	0.067	0.248	0.518

Select the correct option

Reload

- Newton's forward difference formula
- Newton's interpolation formula
- Lagrange's interpolation formula
- Newton's backward difference formula

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MTH603 - Numerical Analysis (Quiz No.3)

Question # 7 of 10 (start time: 05:15:52 PM, 13 August 2022)

Given the following data x:0 1 4 y:2 1 4 Value of first order divided difference y[0,1]is

Select the correct option

- 2
- -1
- -2
- 1

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Question # 4 of 10 (start time: 04:38:55 PM, 13 August 2022)

For the given divide difference table

x	y	$1st D.D$	$2nd D.D$
1	2.2	0.4333	
4	3.5	0.2	−0.0389
7	4.1		

the Newton's divide difference interpolation *formula will be*

Select the correct option

- $y = f(x) = 2.2 + (x - 1)(-0.0389) + (x - 1)((x - 4)(0.4333)$
- $y = f(x) = -0.0389 + (x - 1)(2.2) + (x - 1)((x - 4)(0.4333)$
- $y = f(x) = -0.0389 + (x - 1)(0.4333) + (x - 1)((x - 4)(2.2)$
- $y = f(x) = 2.2 + (x - 1)(0.4333) + (x - 1)((x - 4)(-0.0389)$

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Question # 10 of 10 (Start time: 04:45:37 PM, 13 August 2022) **Total Marks:** 1

For the given data points $(2,5)$, $(4,7)$, *and* $(6,9)$, *the* first $-$ *order divide difference will be*

Select the correct option

Reload Math Equations

- 1
- 0
- 5
- 2

CamScanner

Question # 4 of 10 (Start time: 05:12:42 PM, 13 August 2022) **Total Marks:** 1

For the giev two data points , *the* degree *of Lagrange's* interpolation *polynomial could be*

x	0.3	0.7
y	0.067	0.248

Select the correct option

Reload Math Equations

- ○ one
- ○ three
- ○ two
- ○ four

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Question # 10 of 10 (Start time: 04:03:51 PM, 13 August 2022) **Total Mark**

In Lagrange's interpolation, for n values of y corresponding to n values of x, we can represent the function f (x) by a polynomial of degree

Select the correct option

- n
- n+2
- n-1
- n+1

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Question # 8 of 10 (Start time: 04:10:45 PM, 13 August 2022)

What will be the value of 'a' in the given divide difference table?

x	y	1st D. D	2nd D. D	3rd D. D
2	0.5	0.3	a	
4	1.1	0.3	−0.0125	−0.0021
6	1.7	0.25		
8	2.2			

Select the correct option

- 0.016
- 0.0012
- 0
- 0.00032

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MTH603 – Numerical Analysis (Quiz No.3) Quiz start Tim

Question # 9 of 10 (Start time: 04:11:52 PM 13 August 2022)

What will be the value of 'a' in the given divide difference table?

x	y	1st D. D	2nd D. D	3rd D. D
1	0.4	0.25	0.0375	−0.0104
3	0.9	0.4	a	
5	1.7	0.3		
7	2.3			

Select the correct option

- -0.0343
- -0.02
- -0.0104
- -0.0243

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Question # 2 of 10 (start time: 04:37:23 PM, 13 August 2022)

What will be the value of 'a' in the given divide difference table?

x	y	$1st D.D$	$2nd D.D$	$3rd D.D$
2	0.5	0.3	a	
4	1.1	0.3	−0.0125	−0.0021
6	1.7	0.25		
8	2.2			

Select the correct option

- 0
- 0.0893
- 0.0115
- 0.0612

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Question # 3 of 10 (start time: 04:38:06 PM, 13 August 2022) **Total Marks:** 1

For the given data points $(2, 0.3)$, $(4, 1)$, *and* $(6, 1.2)$ *the divide difference table will be given as*

Select the correct option

Reload Math Equations

○

x	y	$1st D.D$	$2nd D.D$
2	0.3	−0.0625	
4	1	0.1	0.35
6	1.2		

○

x	y	$1st D.D$	$2nd D.D$
2	0.3	0.35	
4	1	0.1	−0.0625
6	1.2		

○

x	y	$1st D.D$	$2nd D.D$
2	0.3	0.1	
4	1	−0.0625	0.35
6	1.2		

○

x	y	$1st D.D$	$2nd D.D$
2	0.3	−0.0625	
4	1	0.35	0.1
6	1.2		

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Question # 5 of 10 (Start time: 04:06:42 PM, 13 August 2022) **Total Marks:**

For the given data points $(2, 0.3)$, $(4, 1)$, *and* $(6, 1.2)$ *the divide difference table will be given as*

Select the correct option

○	x y $1stD.D$ $2ndD.D$ 2 0.3 −0.0625 4 1 0.1 0.35 6 1.2
○	x y $1stD.D$ $2ndD.D$ 2 0.3 −0.0625 4 1 0.35 0.1 6 1.2
○	x y $1stD.D$ $2ndD.D$ 2 0.3 0.1 4 1 −0.0625 0.35 6 1.2
○	x y $1stD.D$ $2ndD.D$ 2 0.3 0.35 4 1 0.1 −0.0625 6 1.2

Question # 3 of 10 (Start time: 04:03:32 PM, 13 August 2022)

At which of the following point the derivatives or slopes the functions $f(x) = x - 2$ and $g(x) = x + 2$ may differ?

Select the correct option

- does not differ for any value of 'x'
- differ for every value of 'x'
- $x = -2$
- $x = 2$

Question # 2 of 10 (Start time: 04:02:40 PM, 13 August 2022)

If f(x) = 2x, then which of the following is will be derivative of f(x) at x = 0.2?

Select the correct option

- 2
- 0.2
- -2
- 0.4

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Question # 5 of 10 (Start time: 06:33:18 PM, 13 August 2022)

Which of the following is the Richardson's Extrapolation limit: $F1(h/2)$ provided that $F(h/2) = F(h) = 1$?

Select the correct option

- 0
- 3
- 4
- 1

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Question # 1 of 10 (Start time: 05:19:41 PM, 13 August 2022)

$$\Delta = - - - -$$

Select the correct option

○	$\frac{E-1}{2}$
○	$E - 1$
○	None
○	$1 - E$

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Question # 6 of 10 (Start time: 06:34:00 PM, 13 August 2022)

Richardson extrapolation method is used to improve the rate of convergence of a .....

Select the correct option

- Sequence
- Series

Question # 1 of 10 (Start time: 08:20:37 PM, 13 August 2022) **Total Marks: 1**

For the given data points (1, 0.3), (3, 1), and (5, 1.2) the divide difference table will be given as

Select the correct option

Reload Math Equations

x	y	1st D.D	2nd D.D
2	0.3	0.35	
4	1	0.1	−0.225
6	1.2		

x	y	1st D.D	2nd D.D
2	0.3	0.35	
4	1	0.1	−0.325
6	1.2		

x	y	1st D.D	2nd D.D
2	0.3	0.35	
4	1	0.1	−0.0625
6	1.2		

x	y	1st D.D	2nd D.D
2	0.3	0.35	
4	1	0.1	−0.125
6	1.2		

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Question # 2 of 10 (Start time: 05:20:17 PM, 13 August 2022)

Given the following data x:0 1 4 8 y:1 1 8 16 Value of 1st order divided difference f[4,8] is

Select the correct option

○	2
○	8
○	6
○	4

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Question # 5 of 10 (Start time: 05:22:55 PM, 13 August 2022)

$\delta = - - - -$

Select the correct option

○	$\frac{E^{\frac{1}{2}} + E^{-\frac{1}{2}}}{2}$
○	None
○	$E^{\frac{1}{2}} - E^{-\frac{1}{2}}$
○	$E^{\frac{1}{2}} + E^{-\frac{1}{2}}$

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Question # 1 of 10 (start time: 05:31:41 PM, 12 August 2022)

Given the following data x: 0 1 2 4 y: 1 1 2 6 Value of 1st order divided difference f[3,4] is

Select the correct option

- 2
- -2
- 1
- 0

Question # 3 of 10 (Start time: 05:32:48 PM, 13 August 2022)

For the following data

x	1	2	5	7	8
y	−5	10	20	22	24

the polynomial of the Lagrangens interpolation could be

Select the correct option

- $\frac{5}{11}x^7 - \frac{41}{7}x^3 + \frac{85}{28}x - \frac{130}{3}$
- $-\frac{11}{240}x^4 + \frac{113}{120}x^3 - \frac{7303}{840}x^2 + \frac{13613}{120}x - \frac{299}{3}$
- $-\frac{1121}{40}x^3 + \frac{41}{7}x^2 - \frac{153}{40}x^2 + \frac{18}{91}x - \frac{3}{10}$
- $x^4 - \frac{1}{16}x^3 + \frac{87}{5}x^4 - \frac{477}{161}x + \frac{3}{5}$

Question # 6 of 10 (Start time: 09:37:34 PM, 13 August 2022)

To take the derivative of f(x) = 2x in the interval [-3,3], which of the following partition of subintervals will be suitable?

Select the correct option

- Equally spaced
- Any arbitrary partition will work
- Union of equally spaced and unequally spaced intervals.
- Unequally spaced

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Question # 5 of 10 (Start time: 06:55:37 PM, 13 August 2022)

For the given data points $(1, -3)$, $(2, 0)$, *and* $(3, 15)$, *the zero − order divide difference will be*

Select the correct option

Reload M

- -1
- -2
- -3
- 0

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Question # 3 of 10 (Start time: 06:34:18 PM, 13 August 2022)

x : 3 7 10 f(x) : -3 10 3 13 Which formula is useful in finding the interpolating polynomial?

Select the correct option

- Newton's forward difference formula
- (vii) None of the given choices
- Newton's backward difference formula
- Lagrange's interpolation formula

Question # 10 of 10 (Start time: 06:58:49 PM, 13 August 2022)

Given the following data x:1 3 8 y:2 4 9 f(3) can be found by using

Select the correct option

○	Newton's backward difference interpolation formula
○	None
○	Lagrange's interpolation formula
○	Newton's forward difference interpolation formula

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Question # 9 of 10 (Start time: 05:26:33 PM, 13 August 2022)

For the given data points $(1, -3)$, $(2, 0)$, *and* $(3, 15)$, *the first − order divide difference will be*

Select the correct option

- 2
- -2
- -3
- 3

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Question # 4 of 10 (Start time: 06:35:02 PM, 13 August 2022) Total Marks: 1

For the given divide difference table

x	y	$1st D.D$	$2nd D.D$
2	2.2	0.325	
6	3.5	0.15	−0.0219
10	4.1		

the Newton's divide difference interpolation *formula will be*

Select the correct option

Report Math Equations

- $y = f(x) = 2.2 + (x-2)(x-6)(0.325) + (x-2)(-0.0219)$
- $y = f(x) = 2.2 + (x-12)(0.325) + (x-6)(x-2)(-0.0219)$
- $y = f(x) = 2.2 + (x-6)(0.325) + (x-12)(x-6)(-0.0219)$
- $y = f(x) = 2.2 + (x-2)(0.325) + (x-2)(x-6)(-0.0219)$

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What will be the value of 'a' in the given divide difference table?

x	y	$1stD.D$	$2ndD.D$	$3rdD.D$
2	0.5	0.3	a	
4	1.1	0.3	−0.0125	−0.0021
6	1.7	0.25		
8	2..2			

Select the correct option

Reload Math Equations

- ○ 0
- ○ 0.0893
- ○ 0.0115
- ○ 0.0612

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Question # 7 of 10 (Start time: 06:56:51 PM, 13 August 2022) **Total Marks:** 1

For the given data points (x_0, y_0), $(x_1 y_1)$, $(x_2 y_2)$, *and* (x_3, y_3) *the first − order divide difference will be given as*

Select the correct option

Reload Math Equations

- $y[x_0, x_1]$
- $y[x_0, x_1, x_2]$
- $y[y_0, y_1, y_2]$
- $y[x_0]$

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Question # 4 of 10 (Start time: 08:32:07 PM, 13 August 2022)

Given the following data x:4 5 7 10 y:46 102 294 346 value of 1st order divided difference f[5 , 7] is

Select the correct option

- 98
- 94
- 91
- 92

Question # 3 of 10 (Start time: 09:48:02 PM, 13 August 2022)

At which of the following point the derivatives or slopes the functions f(x) = x − 2 and g(x) = x + 2 may differ?

Select the correct option

- does not differ for any value of 'x'
- x = -2
- differ for every value of 'x'
- x = 2

Question # 1 of 10 (Start time: 05:43:13 PM, 13 August 2022)

In lagrange's interpolation, for the given five points we can represent the function f (x) by a polynomial of degree

Select the correct option

- 3
- 4
- 5
- 6

Question # 7 of 10 (Start time: 06:34:34 PM, 13 August 2022)

Which of the following is the Richardson's Extrapolation limit: F2(h/4) provided that F1(h/4) = F1(h/2) = 1?

Select the correct option

- 16
- 1
- 15
- -1

Question # 2 of 10 (Start time: 05:44:12 PM, 13 August 2022) **Total Marks: 1**

For the given data points (x_0, y_0), (x_1, y_1), (x_2, y_2), *and* (x_3, y_3) *the second − order divide difference will be given as*

Select the correct option

Report MCQ Question

○	$y[x_0]$
○	$y[x_0, x_1, x_2]$
○	$y[x_0, x_1]$
○	$y[x_0, x_1, x_2, x_3]$

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Question # 6 of 10 (Start time: 06:47:22 PM, 13 August 2023)

If $y(x)$ is approximated by a polynomial $P_n(x)$ of degree n then the error is given by

Select the correct option

- $e(x) = y(x) \div P_n(x)$
- $e(x) = y(x) - P_n(x)$
- $e(x) = y(x) + P_n(x)$
- $e(x) = y(x) \times P_n(x)$

Question # 9 of 10 (Start time: 08:37:01 PM, 13 August 2022)

In the process of Numerical Differentiation, we differentiate an interpolating polynomial in place of ----------.

Select the correct option

- extrapolating polynomial
- actual function
- Lagrange's polynomial
- Newton's Divided Difference Interpolating polynomial

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Question # 5 of 10 (Start time: 09:46:53 PM, 13 August 2022)

To take the derivative of f(x) = 2x in the interval [-3,3], which of the following partition of subintervals will be suitable?

Select the correct option

- Any arbitrary partition will work
- Union of equally spaced and unequally spaced intervals.
- Equally spaced
- Unequally spaced

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Question # 7 of 10 (Start time: 05:48:03 PM, 19 August 2022)

Given the following data x: 2 5 y: 4 10 Value of 1st order divided difference f[2 , 5] is

Select the correct option

- -2
- 2
- 0
- 1

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Question # 9 of 10 (Start time: 05:30:03 PM 13 August 2022)

If $f(x) = 2x^5 - 5x^2 + 9x - 6$, then its ——— derivative is zero for all x.

Select the correct option

- 5th
- 2nd
- 3rd
- 4th

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Question # 2 of 10 (Start time: 06:52:21 PM, 13 August 2022) **Total Marks:** 1

If there are (n+2) values of y corresponding to (n+2) values of x, then we can represent the function f(x) by a polynomial of degree

Select the correct option

Reload Math Equations

- n
- n-1
- n+1
- n+2

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Question # 8 of 10 (Start time: 06:35:49 PM, 13 August 2022)

Two-point formula for the first derivative is defined as

Select the correct option

- y(xi+h)-y(xi-h)/h^2
- y(xi+h)-y(xi-h)/2h
- y(xi+h)+y(xi-h)/2h
- y(xi+h)+y(xi-h)/h^2

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Question # 4 of 10 (Start time: 08:48:49 PM, 13 August 2022)

What will be the value of first order divided difference f[x0]for the following data x:0 1 5 y:2 1 6

Select the correct option

○	2
○	3
○	0
○	1

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Question # 1 of 10 (Start time: 06:51:52 PM, 13 August 2022)

Newton's divided difference interpolation formula is used when the values of the independent variable are

Select the correct option

- ○ Not equally spaced
- ○ Equally spaced
- ○ None
- ○ Constant

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Question # 8 of 10 (Start time: 06:57:30 PM, 13 August 2022)

Richardson extrapolation is method also known as .....

Select the correct option

- Series acceleration method
- Sequence acceleration method

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Question # 3 of 10 (Start time: 06:31:06 PM, 13 August 2022)

Differential operator in terms of forward difference operator is given by

Select the correct option

- $D = \frac{1}{h}(\Delta + \frac{\Delta^2}{2} + \frac{\Delta^3}{3} + \frac{\Delta^4}{4} + \frac{\Delta^5}{5} + \ldots)$
- $D = \frac{1}{h}(\Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - \frac{\Delta^4}{4} + \frac{\Delta^5}{5} - \ldots)$
- $D = \frac{1}{h}(\Delta - \frac{\Delta^2}{2!} + \frac{\Delta^3}{3!} - \frac{\Delta^4}{4!} + \frac{\Delta^5}{5!} - \ldots)$
- $D = \frac{1}{h}(\Delta + \frac{\Delta^2}{2!} + \frac{\Delta^3}{3!} + \frac{\Delta^4}{4!} + \frac{\Delta^5}{5!} + \ldots)$

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Question # 2 of 10 (Start time: 06:30:25 PM, 13 August 2022)

At which of the following point the derivatives or slopes the functions $f(x) = x - 2$ and $g(x) = x + 2$ may differ?

Select the correct option

- differ for every value of 'x'
- does not differ for any value of 'x'
- $x = 2$
- $x = -2$

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Question # 9 of 10 (Start time: 06:58:03 PM, 13 August 2022)

---------- difference is a symmetric function of its arguments.

Select the correct option

- ○ Forward
- ○ Backward
- ○ None
- ○ Divided

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Question # 4 of 10 (Start time: 08:53:21 PM, 13 August 2022) Total Marks: 1

For the given data points $(4, 2.2)$, $(8, 3.5)$, *and* $(12, 4.1)$ *the divide difference table will be given as*

Select the correct option

Reload Math Equations

x	y	$1st D.D$	$2nd D.D$
4	2.2	0.325	
8	3.5	0.15	−0.0108
12	4.1		

x	y	$1st D.D$	$2nd D.D$
4	2.2	0.325	
8	3.5	0.15	−0.065
12	4.1		

x	y	$1st D.D$	$2nd D.D$
4	2.2	0.325	
8	3.5	0.15	−0.098
12	4.1		

x	y	$1st D.D$	$2nd D.D$
4	2.2	0.325	
8	3.5	0.15	−0.0219
12	4.1		

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Question # 6 of 10 (Start time: 06:56:16 PM, 13 August 2022)

1st ordered divided difference formula is defined as

Select the correct option

- $y[x0,x1]=(y1-y0)/(x1+x0)$
- $y[x0,x1]=(y1+y0)/(x1-x0)$
- None of the given choices
- $y[x0,x1]=(y1-y0)/(x1-x0)$

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Question # 10 of 10 (Start time: 05:27:51 PM, 13 August 2022)

Newton's divided difference interpolation formula is used when the values of the independent variable are

Select the correct option

○	Not equally spaced
○	Constant
○	Equally spaced
○	None

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Question # 8 of 10 (Start time: 05:25:36 PM, 13 August 2022)

What will be the value of first order divided difference f[1,5]for the following data x:0 1 5 y:2 1 5

Select the correct option

○	2
○	0
○	1
○	3

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For the given data points (x_0, y_0), $(x_1 y_1)$, $(x_2 y_2)$, and (x_3, y_3) the second − order divide difference will be given as

$$y[x_0, x_1, x_2]$$

$$y[x_0, x_1]$$

$$y[x_0, x_1, x_2, x_3]$$

$$y[x_0]$$

Click to S

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Question # 4 of 10 (Start time: 05:22:05 PM, 13 August 2022)

To take the derivative of $f(x) = 2x$ in the interval $[-3,3]$, which of the following partition of subintervals will be suitable?

Select the correct option

○	Any arbitrary partition will work
○	Unequally spaced
○	Union of equally spaced and unequally spaced intervals.
○	Equally spaced

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Question # 3 of 10 (Start time: 05:21:05 PM, 13 August 2022)

In Richardson's extrapolation method, we usually use two different step sizes .....and ..... to yield a higher order method.

Select the correct option

○	h/2,h/3
○	h/2,h/4
○	h,h/2
○	None of the given choices

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